

Project 5

Actuation-Redundant Planar Parallel Manipulator - RRR

Format: Group of two students; simulation-based project.

Motivation and context

Actuation redundancy allows a robot to avoid singularities that would otherwise limit its workspace and degrade its stiffness and accuracy. Indeed, the extra actuators can be used to control internal forces and maintain a well-conditioned Jacobian even near configurations that would be singular for a non-redundant robot.

A kinematically-redundant planar parallel robot is shown in Fig. 1. The robot is also shown in the online video https://youtu.be/_vp1ELEtDN4. Two 2-PRR legs are attached to an articulated platform by revolute joints. Each leg is able to control the position of its corresponding revolute joint. The articulated end-effector platform has a passive revolute joint. The robot is therefore kinematically redundant: the end-effector has an additional DOF related to the revolute joint.

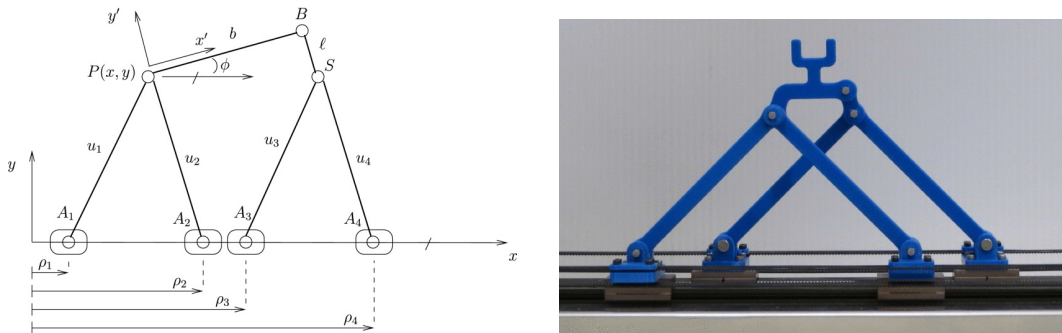


Figure 1: Kinematics architecture and prototype of the kinematically-redundant planar robot [1].

In this project, a similar but **redundantly-actuated** case is considered. The passive revolute joint from the articulated end-effector is removed. Furthermore, the PRR leg mechanisms are replaced by RRR legs. This robot will be referred as 4-RRR. The non-redundant case, i.e., a 3-RRR robot, consists of two RRR legs on one end of the platform and a single RRR leg on the other end. These two robots are shown in Fig. 2.

Tasks

Overall objectives. Analyze the singularities of both 3-RRR and 4-RRR robots. Show that, while the 3-RRR robot cannot undergo unlimited rotation, the 4-RRR robot can do so with an appropriate redundancy-resolution strategy.

- **Compute analytical forward and inverse kinematics.** Is there any difference between non-redundant and actuation-redundant cases?

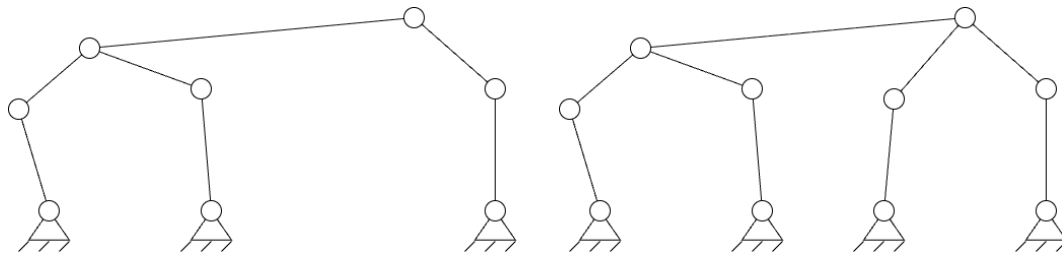


Figure 2: Kinematics architecture of 3-RRR and 4-RRR robots.

- **Analyze the singularities.** Use results from Grassmann geometry to provide a geometrical interpretation.
- **Create a Simscape model of the actuation-redundant robot.** Use this model in order to verify the kinematics equations.
- **Design a trajectory in task space.** This trajectory should illustrate the unlimited rotation capability.
- **Determine joint torques.** Consider quasi-static assumption with mass associated to the end-effector and gravity. Are the simulation results consistent with the singularity analysis?

Available documents

Paper [1] can be accessed online.

Expected outcome

Students should obtain the forward and inverse kinematics and the Jacobian matrices of the actuation-redundant robot. They should understand why and how actuation redundancy allows for doing unlimited rotation.

Submit a report, and a short video (screen recording) with Simscape illustrating unlimited rotation.

References

- [1] C. Gosselin, T. Laliberté, and A. Veillette, “Singularity-free kinematically redundant planar parallel mechanisms with unlimited rotational capability,” in *IEEE Transactions on Robotics*, 2015, doi: 10.1109/TRO.2015.2409433.